

# AMRITANSHU RAI

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[LinkedIn](#) | [GitHub](#) | [rai.codes](#) (portfolio) | [uiready.dev](#) | [odigra.com](#)

## SUMMARY

Frontend engineer with **8+ years** of experience (2018-2026) across augmented reality, digital media and fintech. Most recently **Staff Software Engineer** at Blippar, where I owned client-side media processing, asset pipelines and build tooling for a 200,000-line browser-based **WebAR** studio. Before that I unified 40+ publishing sites at Condé Nast and built the loan-origination journeys for abfldirect.com at Kuliza. I work deep in **React**, **TypeScript** and **WebAssembly**, and I ship the infrastructure my code runs on.

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), TypeScript, HTML, CSS

**Frontend:** React, React Router, Redux, Immutable.js, Tailwind CSS, Material UI, AngularJS, Gatsby

**Backend and data:** Node.js, Express, PostgreSQL, Prisma, MongoDB, Redis, REST, GraphQL, OAuth2

**Browser platform:** WebAssembly, Web Workers, Service Workers, PWA, IndexedDB, WebGL, WebAR, server-side rendering

**Build and testing:** Webpack 5, Vite, Babel, ESLint, Jest, React Testing Library, Playwright

**Infrastructure:** Docker, Kubernetes, AWS, Cloudflare, Drone CI, GitHub Actions, Nginx, Caddy

**Also:** web performance, internationalization, accessibility, SEO, code review, mentoring

## PROFESSIONAL EXPERIENCE

### Blippar (BlippBuilder India Private Limited), Bengaluru

**Staff Software Engineer** | October 2022 - June 2026

Blippbuilder is Blippar's no-code, browser-based **WebAR** authoring platform for publishing augmented reality experiences with marker, surface (SLAM), face and gyro tracking.

- Built a client-side video transcoding pipeline with **FFmpeg.wasm** that compresses uploads in the browser, cutting upload size and server processing before a file leaves the device. Added telemetry for original and final size, bitrate and duration so the savings are measurable in production.
- Sized a **Web Worker** pool from **navigator.hardwareConcurrency** with WASM threading, instance leasing and a promise-backed queue, so several videos transcode in parallel. Warm-up runs after window load and never delays first paint.
- Replaced unreliable video-element probing with **mediainfo.js (WebAssembly)** for accurate metadata, lazy-loaded as its own **Webpack** chunk behind a factory initializer with timeout handling.
- Rebuilt the AR marker manager and added quality scoring that gates publishing on how trackable an image is, cutting tracking failures. Added **HEIC/HEIF** conversion and two-stage compression, with batch uploads that survive single-file errors.
- Rewrote the **Redux** upload queue around an explicit task state machine, removing a class of races where uploads fired before transcoding finished. Shipped an asset library browser that swaps **3D models** and sprite textures while preserving legacy world-unit scaling.
- Led **Webpack 5** migration work, modernizing plugin APIs and WASM handling and tuning code splitting for a 100+ MB bundle. Consolidated hub SSO through authenticated proxies in the **Node.js** and **Express** BFF, and extended **i18next** localization to six languages including Arabic RTL.
- Contributed **150+ commits** to a 1,200-file, 200,000-line **React** and **Redux** codebase built on a micro-kernel with 60 editor plugins, shipping across 8 release trains.

### Condé Nast, Bengaluru

**Software Engineer** | January 2021 - October 2022

Global media company behind Vogue, Wired, Vanity Fair and other leading digital brands.

- Migrated the legacy **Vogue** and **Wired** sites onto Condé Nast's new design system in **React**.
- United **40+ Condé Nast websites** behind a single repository using **React** and **Node.js**, raising code reuse and performance.
- Built an internal **React** tool that let managers control the look and feel of hundreds of components, so minor changes no longer needed an engineer.
- Wrote several **Chrome extensions** in HTML, CSS and React that supported GitHub pull request testing and scraped useful data from production sites.

## Kuliza Technologies, Bengaluru

Frontend Developer | January 2018 - March 2020

Product engineering firm delivering digital lending and fintech applications for enterprise clients.

- Built and maintained **abfdirect.com** and its personal- and business-loan origination journeys in **AngularJS (1.x)**.
- Delivered responsive, cross-browser layouts and integrated **Razorpay** for EMI auto-debit.
- Led **3 frontend developers**, assigning work and conducting code reviews.
- Merged the two loan codebases into one, **halving code size**; raised **Lighthouse performance scores by more than 30%**.

## PROJECTS

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### UIReady, [uiready.dev](#) | Solo Founder and Engineer

2026 - Present

Frontend-interview practice platform with an in-browser IDE, live Jest tests and diagram-backed solutions.

- Built the full stack with **React 19, React Router 7 SSR, TypeScript, Vite, Tailwind CSS v4, Express, Prisma and PostgreSQL 17**: 261 commits, ~24,000 lines of application code and **663 questions**.
- Created a headless **Sandpack** workspace with live Jest runs, a resizable IDE and debounced **Cloudflare R2** autosave; generated **428 React, Vue, Angular and Vanilla TypeScript variants** from shared question definitions.
- Built a validated Vite/MDX/Shiki publishing pipeline for public and private R2 content; added solution pass/fail checks and accessibility validation across **2,835 SVG diagrams**.
- Shipped **763 sitemap URLs**, 83 facet landing pages, JSON-LD/FAQPage schema, llms.txt and dynamic Open Graph images using **Satori and resvg**.
- Integrated **Dodo Payments**, PPP pricing in **74 countries** and Svix-verified idempotent webhooks. Operate Docker on Oracle ARM with Caddy, Cloudflare, monitoring and hourly R2 backups; fixed payment authorization, entitlement and pricing vulnerabilities.

### Odigra, [odigra.com](#) | Solo Founder and Engineer

2026 - Present

Résumé-led job-search extension that crawls listings, filters mismatches locally and ranks opportunities with explanations.

- Built a **Manifest V3 Chrome extension** with a shared Firefox build, PDF/DOCX résumé profiles and adapters for **14 job-board and ATS source families**.
- Implemented a resumable service-worker crawler with **three bounded browser lanes**, persisted checkpoints, chrome.alarms, paced retries and recovery after verification challenges; isolated account/workspace state in **IndexedDB**.
- Applied deterministic relevance, location and work-authorization filters before paid AI evaluation. Built the **PostgreSQL 17** service with AES-GCM-encrypted profiles, forced row-level security, durable worker leases and four-way 100-job evaluation rounds.
- Added scoped OAuth2 credentials, safe exports, DOMPurify sanitization and reproducible esbuild packaging; **189 passing Node.js tests** cover permissions, CSP, account isolation, recovery and Chrome/Firefox parity. Deploy with Docker, Cloudflare and encrypted R2 backups.

### Personal Portfolio, [rai.codes](#)

2019 - Present

Created this in Gatsby.js in 2019 without AI; now migrated to HTML/CSS/JavaScript.

- Removed React/Gatsby and styling runtimes, delivering a static site with **zero package dependencies** and no build step.
- Preserved the responsive design, WebGL animation and offline caching; improved keyboard focus, reduced-motion support and animation lifecycle handling, verified through local UI comparisons and automated checks.

## EDUCATION

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### SRM University, Chennai, Tamil Nadu

Bachelor's in Software Engineering | March 2013 - March 2017